

ASHIM DAHAL

Undergraduate Researcher in Computer Vision & Multimodal Learning
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RESEARCH INTERESTS

3D/4D Computer Vision, Video Understanding, Vision-Language Models, Embodied Perception, Neural Rendering, Efficient Deep Learning

EDUCATION

University of Southern Mississippi

Aug 2023 – May 2027 (Expected)

Bachelor of Science in Computer Science; GPA: 3.87/4.0

Hattiesburg, MS, USA

– Keystone Honors Scholar and merit-based full-tuition scholarship recipient.

PEER-REVIEWED PUBLICATIONS

Journal Articles

- [J1] A. **Dahal**, A. Ghimire, S. A. Murad, and N. Rahimi, “Redemption score: A multi-modal evaluation framework for image captioning via distributional, perceptual, and linguistic signal triangulation,” *IEEE Transactions on Multimedia*, 2026, Accepted; co-first author, **[Impact Factor: 9.9]** [arXiv].
- [J2] S. A. Murad, A. **Dahal**, and N. Rahimi, “Multilingual cyber threat detection in Tweets/X using ML, DL, and LLM: A comparative analysis,” *IEEE Transactions on Computational Social Systems*, vol. 13, no. 2, pp. 1758–1772, 2026, **[Impact Factor: 4.9]** [arXiv].
- [J3] A. **Dahal**, S. A. Murad, and N. Rahimi, “Heuristic comparison of vision transformers against convolutional neural networks for semantic segmentation on remote sensing imagery,” *IEEE Sensors Journal*, vol. 25, no. 10, pp. 17 364–17 373, 2025, **[Impact Factor: 4.5]** [arXiv].

Workshop Papers

- [W1] A. **Dahal**, A. Ghimire, S. A. Murad, and N. Rahimi, “POVQA: Preference-optimized video question answering with rationales for data efficiency,” in *IEEE/CVF (CVPRW)*, Proceedings Track, 2026 [arXiv].
- [W2] A. **Dahal**, S. A. Murad, and N. Rahimi, “Embedding shift dissection on CLIP: Effects of augmentations on VLMs’ representation learning,” in *IEEE/CVF (CVPRW)*, Proceedings Track, 2025 [arXiv].

PREPRINTS

- [P1] A. **Dahal** and B. Banerjee, *Metasampling: Making frame samplers efficient for long-video question answering*, 2026, Submitted to CVPR [arXiv].
- [P2] A. **Dahal**, R. Abdelfattah, and N. Rahimi, *Does it matter which gaussians you pick in 4d gaussian streaming?* 2026, Submitted to CVPR [arXiv].

RESEARCH EXPERIENCE

MCTS-VQA: Monte Carlo Tree Search for Video Evidence Collection [P1] Jan 2026 – Present

Undergraduate Researcher

Advisor: Bikramjit Banerjee

- Developing a Monte Carlo Tree Search framework for identifying and collecting question-relevant frame evidence in long-form video question answering.
- Developed MetaSampling to reduce frame counts by up to 18% while improving accuracy by up to +3.35% [P1].

Anchor Policies for Efficient 4D Gaussian Streaming [P2]

Apr 2025 – Jan 2026

Student Principal Investigator

Advisors: Rabab Abdelfattah and Nick Rahimi

- Developed adaptive anchor-management policies for efficient streaming of dynamic 4D Gaussian representations.
- Investigated anchor insertion, retention, and removal policies to improve memory efficiency while preserving temporal and rendering quality.
- Independently proposed and led the project through a competitive \$5,500 DCUR Summer Research Grant.

Cyber Innovations Lab, USM [J1] [J2] [J3] [W1] [W2]

Aug 2023 – Present

*Undergraduate Research Assistant (fully funded); Advisor: Nick Rahimi**Hattiesburg, MS, USA*

- **POVQA**: Developed a rationale-guided video QA framework that reduced input-token use by 40% while retaining over 95% of baseline accuracy. [W1]
- **Redemption Score**: Developed a multimodal image-caption evaluation framework combining distributional, perceptual, and linguistic signals for robust image-text alignment assessment. [J1]
- **CLIP Representation Analysis**: Led controlled ablation studies measuring embedding shifts caused by image augmentations in CLIP and related vision-language models. [W2]
- **Remote-Sensing Segmentation**: Compared Vision Transformers and convolutional neural networks for semantic segmentation on satellite imagery and developed a weighted loss function for imbalanced classes. [J3]
- **NASA EPSCoR Project**: Co-designed the research methodology for a \$51,000 NASA EPSCoR-funded project on efficient vision-language models, including objectives, evaluation protocols, and deployment strategies.

HONORS & AWARDS

\$1,500 Eagle SPUR Research Grant – Drapeau Center for Undergraduate Research	2026
\$500 Best Presentation on Computational Approaches – Drapeau Center for Undergraduate Research	2026
\$5,500 Summer Research Grant – Drapeau Center for Undergraduate Research	2025
\$51,000 NASA EPSCoR Project – Research Team Member & Technical Proposal Co-Drafter	2025 – 2026
Keystone Honors Scholar – University of Southern Mississippi	2025 – 2027
\$500 Hatchery Checkpoint Award	2024
\$200 Eagles Write Award – School of Humanities, USM	2024

LEADERSHIP & SERVICE

Lead Organizer	2025 – Present
<i>Google Developer Groups on Campus at University of Southern Mississippi</i>	<i>Hattiesburg, MS</i>
– Organize GDG seminars, a Robo Soccer tournament, and twice-per-semester all-community meetings.	
– Represented USM at Google I/O in Sunnyvale, California, in May 2026 as a sponsored delegate.	
Research Liaison	2025 – Present
<i>School of Computing Sciences and Computer Engineering Student Ambassadors</i>	<i>Hattiesburg, MS</i>
– Represent undergraduate research interests and connect prospective students with research opportunities and faculty mentors.	
Head of Artificial Intelligence	2024 – 2025
<i>Google Developer Student Club at University of Southern Mississippi</i>	<i>Hattiesburg, MS</i>
– Delivered the “What Is an Image?” workshop on digital images, color spaces, and image processing to 60+ undergraduate students.	
– Led AI/ML workshops and mentored 20+ students on machine-learning projects and research directions.	

TEACHING & MENTORSHIP EXPERIENCE

Machine Learning Bootcamp Series	2022 – 2023
<i>Lead Instructor, Data Research Council for Students</i>	<i>Nepal</i>
– Designed and taught multi-day Python and machine-learning programs covering programming fundamentals, CNNs, PyTorch, and hands-on projects.	
Robotics Instructor	2021 – 2022
<i>Durbar High School</i>	<i>Kathmandu, Nepal</i>
– Designed and taught robotics lessons covering Arduino programming, sensor integration, and mechanical design.	

TECHNICAL SKILLS

Programming : Python, C++, CUDA, Bash, SQL, JavaScript
Machine Learning : PyTorch (DDP, FSDP, Lightning), Hugging Face Transformers, Accelerate, scikit-learn, XGBoost
Computer Vision : 3D/4D Gaussian Splatting, NeRFs, Vision Transformers, CLIP, video understanding, image segmentation, multiview stereo
Systems & Tools : SLURM/HPC, Docker, Git/GitHub, Weights & Biases, NumPy, Pandas, OpenCV, Linux, L ^A T _E X